****
When publishing work with results generated by this powder diffractometer, please include the following sentence in the Acknowledgements section:
'The support of the National Science Foundation (CHE-1429768) for the purchase of the powder X-ray diffractometer is gratefully acknowledged.'
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Grant Reporting Statistics

Installation of XRD data analysis software

For both the D2 and D8 we use Diffrac.eva to analyze data. When you install diffrac.eva the installer will install 3 items: codemeter (allows you to connect to the license server, local access only - you can't run it from home), diffrac.eva, and the COD (crystallography open database). Sometimes for unknown reasons the COD does not install, so install that separately after running the main install. ***Note these are Windows only programs and diffrac.eva will only work on a computer on the chemistry network - requires a license checkout from the license server***.

diffrac.eva installer

COD Installer

diffrac.eva Manual

diffrac.eva Tutorial

diffrac.eva Cluster Analysis Manual

D2 Phaser Info and SOPs

D2 General SOP

D2 Performance

D2 Training Sheet

D2 Daily Checks

D2 Daily Log

D2 2-Theta Accuracy Check

Bruker supplied training videos for the D2 Phaser

D8 Advance Info and SOPs

D8 General SOP

D8 Rotating Stage SOP

D8 Capillary Stage SOP

D8 Variable Temperature Stage SOP

D8 Performance

D8 Training Sheet

D8 Daily Checks

D8 Daily Log

D8 XYZ Stage SOP

UNR EH&S Manuals

Radiation Safety Manual

X-Ray and Accelerator Radiation Safety Program

Link to UNR's radiation safety webpage

D8 Advance XRD general SOP

5/22/2018 Lab manager: Janina Ruprecht jruprecht@unr.edu Cell 917 476 5641

For more information on the XRD instruments please visit http://argenta2.rd.unr.edu/x-ray-powder-diffractometer.html

Turning on the D8 (with desired stage already attached)

1) Check if the sample stage that suits your needs is attached to the goniometer. If not, and you had no training on how to exchange them, please ask the system manager to swap them out for you. This may take one hour or more and may require LN2 if using the temperature stage, so please plan ahead

2) Prepare sample in holder suitable for attached sample stage; check the SOP of the used sample stage for instructions

3) Start the cooling cycle:
   a. Open up the two water mains valves on the wall between the desks (handles have to line up with the lines)
   b. Switch on the Haskris chiller behind the D8 PC

4) Turn on the instrument:
   a. On the lower left-hand side of the instrument, turn the large yellow-and-red Mains knob from O to I, wait 15 seconds
   b. Press the green "Power-on" button next to the Mains knob
   c. Wait for the "I" to appear on the top button on the left-hand front of the instrument (generator button, takes roughly 50 seconds)
   d. Once "I" has appeared press that same generator button; it will start blinking yellow and after 3-4 seconds it should become a black on yellow radiation symbol; when cathode is fully heated up the symbol will turn to yellow on black

***NOTE**: After 2 days powered down then the instrument will do a 45 minute tube conditioning (generator button flashing blue with "COND" on it). When the instrument is finished doing the tube conditioning, the generator button will become a yellow on black radiation symbol*

5) Start the computer:
   a. Turn on the computer – ask Nina or Stephen for password
   b. Launch DIFFRAC.measurement from bottom task bar, log in using Bruker Service with password "Bruker-Service"
   c. After a while you should see the Measurement Server starting. Open it (task bar, right side), hit "Connect", then close window

***NOTE:** If this is the first launch after everything has been off it will take a while for everything to connect*

Scan settings for D8 and managing your results

NOTE: *The numbers given below are just standard settings and you're free to change them to what you need. The numbers that you should* ***NOT*** *mess with are the x-ray power and current as well as the STEPS parameter.*

1) Choose primary (X-ray tube) and secondary (detector) optics in "Commander":

   a. In DIFFRAC.measurement, go to the "Commander" tab and wait until everything is connected

   b. Setup if using the **motorized slits**:
      - Twin Primary: Motorized Slit: Slit width 0.681. You must hit the "Apply new values" button to activate the setting
      - Twin Secondary: Motorized Slit: Slit width 0.681. You must hit the "Apply new values" button to activate the setting
      - **Insert the Nickel K_β** filter on the secondary side of the optics

   c. Setup if using the **Göbel mirrors**:
      - Twin Primary: The Göbel mirror is more forgiving of poorly ground/defined samples (seems to give better S/N)
      - Twin Secondary: Motorized Slit: Slit width wide open at 5.4 mm. You must hit the "Apply to settings" button to activate the setting
      - **Insert a fixed slit** (standard is 0.1 mm) on the primary side or your peaks will be broad, and **remove the $K_\beta$filter** on the secondary side since the K_β radiation is already removed by the Göbel mirror

2) Initialize drives and x-ray tube:
   c. In "Commander" in the top left panel, anything with a caution symbol needs to be initialized. Click the checkbox to the right of each drive value and then click the "Initialized all checked drives" button, agree "yes"
   d. In the lower X-ray generator section, set Voltage to 40 kV and Current to 40 mA, hit the SET button

The system is ready for scanning

3) Choose scan parameters and start scan:

   a. If not done already, Initialize drives and x-ray tube (see above)
   b. In the “Scan setup” panel, fill in desired scan settings (see below for general settings); click into a white box several times to update the Effective total time [sec]; adjust dwell time and increments if total time does not fit your needs
   c. Hit START; the X-rays will turn on, power is ramped up, drives go to starting position and the scan is started
   d. **After each scan, save your results file or the data will be lost!** When the scan has finished go to FILE > SAVE RESULTS ***both*** as .brml and .raw in a local folder and only then prepare next scan or sample

General Settings for Data Acquisition:

Sample rotation:	15 rpm
Time (s):	Your choice; more time equals better signal/noise (S/N)
2Theta angles:	Your choice; depends on your sample type
Increment (°):	Your choice; a smaller increment equals better S/N but you should set the increments so you get at least 5 data point above the half-height
Scan type:	Coupled two theta/theta
Scan mode:	Continuous PSD fast (Position Sensitive Detector)
PSD opening:	4.84876814600

NOTE: *Do* ***NOT*** *adjust the Steps parameter, it is automatically calculated from the other parameters. By adjusting steps improperly you can crash the optics!*

D8 Advance XRD with Rotating stage SOP

5/21/2018 Lab manager: Janina Ruprecht jruprecht@unr.edu Cell 917 476 5641

For more information on the XRD instruments please visit http://argenta2.rd.unr.edu/x-ray-powder-diffractometer.html

Turning on the D8 (with desired stage already attached)

1) Start the cooling cycle:
   a. Open up the two water mains valves on the wall between the desks (handles have to line up with the lines)
   b. Switch on the Haskris chiller behind the D8 PC

2) Turn on the instrument:
   a. On the lower left-hand side of the instrument, turn the large yellow-and-red Mains knob from O to I, wait 15 seconds
   b. Press the green "Power-on" button next to the Mains knob
   c. Wait for the "I" to appear on the top button on the left-hand front of the instrument (generator button, takes roughly 50 seconds)
   d. Once "I" has appeared press that same generator button; it will start blinking yellow and after 3-4 seconds it should become a black on yellow radiation symbol; when cathode is fully heated up the symbol will turn to yellow on black

***NOTE**: After 2 days powered down then the instrument will do a 45 minute tube conditioning (generator button flashing blue with "COND" on it). When the instrument is finished doing the tube conditioning, the generator button will become a yellow on black radiation symbol*

3) Start the computer:
   a. Turn on the computer – ask Nina or Stephen for password
   b. Launch DIFFRAC.measurement from bottom task bar, log in using Bruker Service with password "Bruker-Service"
   c. After a while you should see the Measurement Server starting. Open it (task bar, right side), hit "Connect", then close window

***NOTE:** If this is the first launch after everything has been off it will take a while for everything to connect*

Preparing the D8 for scan with rotating stage

1) Prepare sample in the rotating stage sample holder:
   a. To open the door to the XRD push the door open button ->||<- on front right-hand side of instrument; even if the x-rays are on the shutter will shield them from exiting the primary optics (check the LEDs on the x-ray tube for shutter status, if they are bright red the shutter is open)
   b. The rotation stage has a sample holder that is magnetically attached, simply pull down to remove it
   c. Load/unload samples in the magnetic metal holder ***OUTSIDE*** of the XRD cabinet. The holder is spring loaded and we do not want any sample spill inside the XRD cabinet. Make sure the sample surface is flush with the surface of the holder, otherwise your peaks will be shifted
   d. Re-insert the rotating stage sample holder slowly but with a firm grip, it will snap in place and if you let it snap without a firm grip your powder sample could move around
   e. Adjust the knife air scatter screen positioned above the sample surface to properly remove background air scattering. Using the thumbwheel near the top, move blade up/down according to incident angles used (3mm above sample surface is standard; lower it for very low angle scans)

2) Choose primary (X-ray tube) and secondary (detector) optics in "Commander":

   a. In DIFFRAC.measurement, go to the "Commander" tab and wait until everything is connected

   b. Setup if using the **motorized slits**:
      - Twin Primary: Motorized Slit: Slit width 0.681. You must hit the "Apply new values" button to activate the setting
      - Twin Secondary: Motorized Slit: Slit width 0.681. You must hit the "Apply new values" button to activate the setting
      - **Insert the Nickel K_β** filter on the secondary side of the optics

   c. Setup if using the **Göbel mirrors**:
      - Twin Primary: The Göbel mirror is more forgiving of poorly ground/defined samples (seems to give better S/N)
      - Twin Secondary: Motorized Slit: Slit width wide open at 5.4 mm. You must hit the "Apply to settings" button to activate the setting
      - **Insert a fixed slit** (standard is 0.1 mm) on the primary side or your peaks will be broad, and **remove the $K_\beta$filter** on the secondary side since the K_β radiation is already removed by the Göbel mirror

4) Initialize drives and x-ray tube:
    c. In “Commander” in the top left panel, anything with a caution symbol needs to be initialized. Click the checkbox to the right of each drive value and then click the “Initialized all checked drives” button, agree “yes”
    d. In the lower X-ray generator section, set Voltage to 40 kV and Current to 40 mA, hit the SET button

The system is ready for scanning

NOTE: *After 5 min in idle the system will ramp down the Voltage and Current to 20 kV and 5 mA*

Scan settings for D8 and managing your results

NOTE: *The numbers given below are just standard settings and you're free to change them to what you need. The numbers that you should* ***NOT*** *mess with are the x-ray power and current as well as the STEPS parameter.*

1) Choose scan parameters and start scan:
    a. If not done already, Initialize drives and x-ray tube (see above)
    b. Start sample rotation manually before starting the scan by typing in the desired speed (standard is 15 rpm) and hitting the “Apply new values” button right next to it; Phi should start to rotate
    c. In the “Scan setup” panel, fill in desired scan settings (see below for general settings); click into a white box several times to update the Effective total time [sec]; adjust dwell time and increments if total time does not fit your needs
    d. Hit START; the X-rays will turn on, power is ramped up, drives go to starting position and the scan is started
    e. **After each scan, save your results file or the data will be lost!** When the scan has finished go to FILE > SAVE RESULTS ***both*** as .brml and .raw in a local folder and only then prepare next scan or sample

General Settings for Data Acquisition:

Sample rotation:	15 rpm
Time (s):	Your choice; more time equals better signal/noise (S/N)
2Theta angles:	Your choice; depends on your sample type
Increment (°):	Your choice; a smaller increment equals better S/N but you should set the increments so you get at least 5 data point above the half-height
Scan type:	Coupled two theta/theta
Scan mode:	Continuous PSD fast (Position Sensitive Detector)
PSD opening:	4.84876814600

NOTE: *Do* ***NOT*** *adjust the Steps parameter, it is automatically calculated from the other parameters. By adjusting steps improperly you can crash the optics!*

Ending your session and shutting down the D8

NOTE: *The .brml and .raw files that you have saved so far can only be opened with DIFFRAC.EVA and JADE; please follow the steps below to make your data accessible to you*

1) Data processing:
   a. Close DIFFRACT.measurement
   b. Import your .brml files in DIFFRAC.EVA and then export them as .xy file; this file extension can be opened in notepad, it will have two columns, 2Theta and CPS, which can be copied to excel.
   c. In DIFFRAC.EVA you can also search/match and append the matches if you like, and then print your spectrum.
   d. You can install DIFFRAC.EVA on your windows PC, just follow the instructions on the argenta2 XRD webpage (see link on top of first page); note that this software can only be used within the chemistry domain network
   e. If you want to refine your scans in Jade please ask Nina or Stephen to arrange Jade usage. The .raw file will be needed for that purpose.
   f. Copy saved files into your argenta2 folder (file browser Y: data \\argenta2.rd.unr.edu > Bruker D2) or on a thumb drive
   g. Log off from Windows

2) Turn off the instrument:
   a. Press the generator button on the top left-hand front of the instrument (after a few seconds the button icon will change from a radiation symbol to a white "I")
   b. Press the white "Stand by" button in the insert on the lower left hand side of the instrument
   c. Turn the big yellow-red Mains knob from I to O (next to the Stand by button)

3) Stop the cooling cycle:
   a. Switch off the Haskris chiller
   b. Close water mains (2 valves) on wall; handles should be perpendicular to lines

Keep UPS PowerVar ON except for UPS maintenance

Removing the Rotating Stage

1) Before switching stages make sure the goniometer arms are out of the way by reinitializing the drives (Commander, top checkbox, Initialized all checked drives)
2) Shut down the computer:
   a. Exit DIFFRAC.measurement (some settings in “Commander” tab will be lost)
   b. Open Measurement Server (task bar, right side), hit “Release control”, then close window
   c. Shut down computer

3) Turn off the instrument:
   d. Press the generator button on the top left-hand front of the instrument (after a few sec the button will change from radiation symbol to a white “I”)
   e. Press the white “Stand by” button on the lower left hand side of instrument
   f. Turn the Mains power knob from I to O (next to the stand by button)

!!! NOTE: You must turn off the Mains power to the instrument before exchanging a stage, if not you will damage the driver board !!!

4) Remove the rotating stage:
   a. To open the door to the XRD push the door open button ->||<- on front right-hand side of instrument; even if the x-rays are on the shutter will shield them from exiting the primary optics (check the LEDs on the x-ray tube for shutter status, if they are bright red the shutter is open)
   b. Remove the knife edge (air scatter) and place the sheath on
   c. The rotation stage has a sample holder that is magnetically attached, just pull down to remove it
   d. Using a T25 wrench remove the outer 3 retaining screws on the goniometer (twelve o’clock, four 0’oclock, and eight o’clock positions): ***note*** the screw at twelve o’clock is longer than the other two. Place the screws in the bottled labeled “***Rotating Stage Screws***”
   e. ***NOTE:*** If attaching the rotation stage you will have to tighten the centering screw (on the goniometer flange to the left of the center) before tightening the three screws, and then release the centering screw again
   f. Turn the whole stage counter-clockwise 45 degrees and pull straight out
   g. Disconnect the driver cable (beige cable head) on the back of the rotating stage using a small flathead screwdriver and then place the rotating stage somewhere inside the instrument but out of the way (I used left side)
   h. The driver cable for the rotating stage needs to be removed from the center of the goniometer. Using your left hand feed the cable through the hole in the goniometer and give it to your right hand then move the cable to the front of the cabinet

The rotating stage is now considered completely removed from the instrument. The status of the stage can be viewed in DIFFRAC.measurement > DAVINCI tab

Ending your session and shutting down the D8

NOTE: *The .brml and .raw files that you have saved so far can only be opened with DIFFRAC.EVA and JADE; please follow the steps below to make your data accessible to you*

1) Data processing:
   a. Close DIFFRACT.measurement
   b. Import your .brml files in DIFFRAC.EVA and then export them as .xy file; this file extension can be opened in notepad, it will have two columns, 2Theta and CPS, which can be copied to excel.
   c. In DIFFRAC.EVA you can also search/match and append the matches if you like, and then print your spectrum.
   d. You can install DIFFRAC.EVA on your windows PC, just follow the instructions on the argenta2 XRD webpage (see link on top of first page); note that this software can only be used within the chemistry domain network
   e. If you want to refine your scans in Jade please ask Nina or Stephen to arrange Jade usage. The .raw file will be needed for that purpose.
   f. Copy saved files into your argenta2 folder (file browser Y: data \\argenta2.rd.unr.edu > Bruker D2) or on a thumb drive
   g. Log off from Windows

2) Turn off the instrument:
   a. Press the generator button on the top left-hand front of the instrument (after a few seconds the button icon will change from a radiation symbol to a white "I")
   b. Press the white "Stand by" button in the insert on the lower left hand side of the instrument
   c. Turn the big yellow-red Mains knob from I to O (next to the Stand by button)

3) Stop the cooling cycle:
   a. Switch off the Haskris chiller
   b. Close water mains (2 valves) on wall; handles should be perpendicular to lines

Keep UPS PowerVar ON except for UPS maintenance

D8 Advance XRD with Capillary Stage SOP

5/22/2018 Lab manager: Janina Ruprecht jruprecht@unr.edu Cell 917 476 5641

For more information on the XRD instruments please visit http://argenta2.rd.unr.edu/x-ray-powder-diffractometer.html

Turning on the D8 (with desired stage already attached)

1) Start the cooling cycle:
   a. Open up the two water mains valves on the wall between the desks (handles have to line up with the lines)
   b. Switch on the Haskris chiller behind the D8 PC

2) Turn on the instrument:
   a. On the lower left-hand side of the instrument, turn the large yellow-and-red Mains knob from O to I, wait 15 seconds
   b. Press the green "Power-on" button next to the Mains knob
   c. Wait for the "I" to appear on the top button on the left-hand front of the instrument (generator button, takes roughly 50 seconds)
   d. Once "I" has appeared press that same generator button; it will start blinking yellow and after 3-4 seconds it should become a black on yellow radiation symbol; when cathode is fully heated up the symbol will turn to yellow on black

***NOTE**: After 2 days powered down then the instrument will do a 45 minute tube conditioning (generator button flashing blue with "COND" on it). When the instrument is finished doing the tube conditioning, the generator button will become a yellow on black radiation symbol*

3) Start the computer:
   a. Turn on the computer – ask Nina or Stephen for password
   b. Launch DIFFRAC.measurement from bottom task bar, log in using Bruker Service with password "Bruker-Service"
   c. After a while you should see the Measurement Server starting. Open it (task bar, right side), hit "Connect", then close window

***NOTE:** If this is the first launch after everything has been off it will take a while for everything to connect*

Preparing the D8 for scan with capillary stage

1) Prepare sample in capillary sample holder:
   a. To open the door to the XRD push the door open button ->||<- on front right-hand side of instrument; even if the x-rays are on the shutter will shield them from exiting the primary optics (check the LEDs on the x-ray tube for shutter status, if they are bright red the shutter is open)

b. Load/unload sample capillaries on the capillary goniometer attachment outside of the XRD cabinet. We don't want any sample spill inside the XRD cabinet, and you have to align it anyways
c. Locate the capillary attachment. Keep it on the black holder in order to make loading easier.
d. Fill a glass capillary with at least 3cm of sample, making sure that there are no large gaps. If gaps appear, try tapping or flicking GENTLY.
e. Break off top part of capillary, leaving about 3cm of sample-filled capillary which can be sealed with provided wax. Warm wax first in your hands to make it pliable. By using the wax, attach capillary onto capillary attachment. It has to be centered and pointing out as straight as possible.
f. Attach capillary attachment to alignment tool and align capillary to be as straight as possible and rotating in the center by using the small tool provided. The two screw sets closer to the base of the attachment simply move the capillary in x/y direction, the two screw sets closer to the capillary rotate the capillary to straighten it out. Try centering the base first with the moving screws, then rotate with the rotating screws, and center again. Repeat until capillary appears straight and centered.
g. Attach the capillary attachment onto the stage flange by aligning the little notch on the flange with the gap on the attachment, then push in and tighten the large screw
h. Attach the two air scatter knife edges from above and below. Move upper knife up as far as possible to allow x-ray to hit the capillary

2) Choose primary (X-ray tube) and secondary (detector) optics in "Commander":

a. In DIFFRAC.measurement, go to the "Commander" tab and wait until everything is connected

b. Setup if using the **motorized slits**:
   - Twin Primary: Motorized Slit: Slit width 0.681. You must hit the "Apply new values" button to activate the setting
   - Twin Secondary: Motorized Slit: Slit width 0.681. You must hit the "Apply new values" button to activate the setting
   - **Insert the Nickel K_β** filter on the secondary side of the optics

c. Setup if using the **Göbel mirrors**:
   - Twin Primary: The Göbel mirror is more forgiving of poorly ground/defined samples (seems to give better S/N)
   - Twin Secondary: Motorized Slit: Slit width wide open at 5.4 mm. You must hit the "Apply to settings" button to activate the setting
   - **Insert a fixed slit** (standard is 0.1 mm) on the primary side or your peaks will be broad, and **remove the $K_\beta$filter** on the secondary side since the K_β radiation is already removed by the Göbel mirror

4) Initialize drives and x-ray tube:
   c. In “Commander” in the top left panel, anything with a caution symbol needs to be initialized. Click the checkbox to the right of each drive value and then click the “Initialized all checked drives” button, agree “yes”
   d. In the lower X-ray generator section, set Voltage to 40 kV and Current to 40 mA, hit the SET button

The system is ready for scanning

NOTE: *After 5 min in idle the system will ramp down the Voltage and Current to 20 kV and 5 mA*

General settings for the capillary stage

NOTE: *The numbers given below are just standard settings and you're free to change them to what you need. The numbers that you should* ***NOT*** *mess with are the x-ray power and current as well as the STEPS parameter.*

1) Choose scan parameters and start scan:
   a. If not done already, Initialize drives and x-ray tube (see above)
   b. Start sample rotation manually before starting the scan by typing in the desired speed (standard is 15 rpm) and hitting the “Apply new values” button right next to it; Phi should start to rotate
   c. In the “Scan setup” panel, fill in desired scan settings (see below for general settings); click into a white box several times to update the Effective total time [sec]; adjust dwell time and increments if total time does not fit your needs
   d. Hit START; the X-rays will turn on, power is ramped up, drives go to starting position and the scan is started
   e. **After each scan, save your results file or the data will be lost!** When the scan has finished go to FILE > SAVE RESULTS ***both*** as .brml and .raw in a local folder and only then prepare next scan or sample

General Settings for Data Acquisition:

Sample rotation:	15 rpm
Time (s):	Your choice; more time equals better signal/noise (S/N)
2Theta angles:	Your choice; depends on your sample type
Increment (°):	Your choice; a smaller increment equals better S/N but you should set the increments so you get at least 5 data point above the half-height
Scan type:	Coupled two theta/theta
Scan mode:	Continuous PSD fast (Position Sensitive Detector)
PSD opening:	4.84876814600

NOTE: *Do* ***NOT*** *adjust the Steps parameter, it is automatically calculated from the other parameters. By adjusting steps improperly you can crash the optics!*

Ending your session and shutting down the D8

NOTE: *The .brml and .raw files that you have saved so far can only be opened with DIFFRAC.EVA and JADE; please follow the steps below to make your data accessible to you*

1) Data processing:
   a. Close DIFFRACT.measurement
   b. Import your .brml files in DIFFRAC.EVA and then export them as .xy file; this file extension can be opened in notepad, it will have two columns, 2Theta and CPS, which can be copied to excel.
   c. In DIFFRAC.EVA you can also search/match and append the matches if you like, and then print your spectrum.
   d. You can install DIFFRAC.EVA on your windows PC, just follow the instructions on the argenta2 XRD webpage (see link on top of first page); note that this software can only be used within the chemistry domain network
   e. If you want to refine your scans in Jade please ask Nina or Stephen to arrange Jade usage. The .raw file will be needed for that purpose.
   f. Copy saved files into your argenta2 folder (file browser Y: data \\argenta2.rd.unr.edu > Bruker D2) or on a thumb drive
   g. Log off from Windows

2) Turn off the instrument:
   a. Press the generator button on the top left-hand front of the instrument (after a few seconds the button icon will change from a radiation symbol to a white "I")
   b. Press the white "Stand by" button in the insert on the lower left hand side of the instrument
   c. Turn the big yellow-red Mains knob from I to O (next to the Stand by button)

3) Stop the cooling cycle:
   a. Switch off the Haskris chiller
   b. Close water mains (2 valves) on wall; handles should be perpendicular to lines

Keep UPS PowerVar ON except for UPS maintenance

Attaching the Capillary Stage

1) Before switching stages make sure the goniometer arms are out of the way by reinitializing the drives (Commander, top checkbox, Initialized all checked drives)

2) Shut down the computer:
   a. Exit DIFFRAC.measurement (some settings in “Commander” tab will be lost)
   b. Open Measurement Server (task bar, right side), hit “Release control”, then close window
   c. Shut down computer

3) Turn off the instrument:
   d. Press the generator button on the top left-hand front of the instrument (after a few sec the button will change from radiation symbol to a white “I”)
   e. Press the white “Stand by” button on the lower left hand side of instrument
   f. Turn the Mains power knob from I to O (next to the stand by button)

!!! NOTE: You must turn off the Mains power to the instrument before exchanging a stage, if not you will damage the driver board !!!

4) Attach the capillary stage:
   a. To open the door to the XRD push the door open button ->||<- on front right-hand side of instrument; even if the x-rays are on the shutter will shield them from exiting the primary optics (check the LEDs on the x-ray tube for shutter status, if they are bright red the shutter is open)
   b. Make sure no cables are sticking out of the center of the goniometer. If so, feed them backwards and out to the right
   c. Align the red dots on top of the goniometer and on the capillary stage goniometer flange, push goniometer flange in, turn whole flange clockwise 45 degrees
   d. Tighten the single centering screw on the goniometer flange to the left of the center; this centers the stage to the goniometer
   e. Using a T25 wrench replace the outer 3 retaining screws on the goniometer (twelve o’clock, four 0’oclock, and eight o’clock positions): ***note*** the screw at twelve o’clock is longer than the other two
   f. Release the centering screw
   g. Attach the power cable

The capillary stage is now fully attached. The status of the stage can be viewed in DIFFRAC.measurement > DAVINCI tab

D8 Advance XRD with MTC-LOWTEMP Temperature chamber SOP

5/21/2018 Lab manager: Janina Ruprecht jruprecht@unr.edu Cell 917 476 5641

For more information on the XRD instruments please visit http://argenta2.rd.unr.edu/x-ray-powder-diffractometer.html

Temperature range: -180° to +450° C

For temperatures < 25° C or temperatures > 300° C vacuum is required

For temperatures < 25° C cooling with LN_2 is required

Turning on the D8 (with desired stage already attached)

1) Start the cooling cycle:
   a. Open up the two water mains valves on the wall between the desks (handles have to line up with the lines)
   b. Switch on the Haskris chiller behind the D8 PC

2) Turn on the instrument:
   a. On the lower left-hand side of the instrument, turn the large yellow-and-red Mains knob from O to I, wait 15 seconds
   b. Press the green "Power-on" button next to the Mains knob
   c. Wait for the "I" to appear on the top button on the left-hand front of the instrument (generator button, takes roughly 50 seconds)
   d. Once "I" has appeared press that same generator button; it will start blinking yellow and after 3-4 seconds it should become a black on yellow radiation symbol; when cathode is fully heated up the symbol will turn to yellow on black

***NOTE**: After 2 days powered down then the instrument will do a 45 minute tube conditioning (generator button flashing blue with "COND" on it). When the instrument is finished doing the tube conditioning, the generator button will become a yellow on black radiation symbol*

3) Start the computer:
   a. Turn on the computer – ask Nina or Stephen for password
   b. Launch DIFFRAC.measurement from bottom task bar, log in using Bruker Service with password "Bruker-Service"
   c. After a while you should see the Measurement Server starting. Open it (task bar, right side), hit "Connect", then close window

***NOTE:** If this is the first launch after everything has been off it will take a while for everything to connect*

Preparing the D8 for scan with temperature stage

1) Prepare sample in Temp chamber sample holder:
   a. To open the door to the XRD push the door open button ->||<- on front right-hand side of instrument; even if the x-rays are on the shutter will shield them from exiting the primary optics (check the LEDs on the x-ray tube for shutter status, if they are bright red the shutter is open)
   b. Vent the sample chamber by cracking open the large vacuum flange, then ***SLOWLY*** crack open the black vacuum break knob (back left hand side of stage; if you do this too quickly your sample will be blown throughout the entire sample chamber), wait until hissing subsides and vacuum flange is easily removed
   c. Load/unload samples in the sample holder ***OUTSIDE*** of the XRD cabinet. We don't want any sample spill inside the stage
   d. Using the Wera 118064 micro screwdriver (black handle with green markings) loosen the thermocouple screw and remove the thermocouple
   e. Loosen two screws at the bottom of the sample holder with a 2.5 mm wrench, pull out the sample holder while pulling down on the cold finger
   f. Use a square of yellow post-it's to stand the sample holder upright, add sample to the sample holder and level out with a glass slide
   g. Place the sample holder back into the instrument stage by pulling down on the cold finger, tighten the two screws at the bottom of the sample holder with a 2.5 mm wrench
   h. Reinsert and tightened the thermocouple using the Wera micro screwdriver
   i. replace the vacuum flange and close it tight
   j. Make sure the Haskris chiller is on, then turn on source and sink water valves for the stage, 2 yellow water valves on the floor behind the instrument. Make sure the pressure is set to 24 psi (pressure gauge on water hose going to instrument)
   k. Turn on Eurotherm: Inside lower front console panel, press white button followed by green button; check in Commander if Temperature control is available in the panel below the X-ray generator panel

2) Set up vacuum (T < 25° C or > 300° C, or if desired) and cooling (T <25° C) if desired:
   a. Close both black vacuum break knobs (back left and right hand side of the stage, both pointing sideways when closed)
   b. Turn on vacuum pump
   c. Turn on vacuum gauge
   d. For cooling < 25° make sure there is at least 20 lbs of liquid nitrogen in the dewar
   e. Turn on compressed air (up on the wall above the Haskris chiller)
   f. Adjust compressed air pressure regulator to 0.5 bar

3) Choose primary (X-ray tube) and secondary (detector) optics in “Commander”:

   a. In DIFFRAC.measurement, go to the “Commander” tab and wait until everything is connected

   b. Setup if using the **motorized slits**:
      - Twin Primary: Motorized Slit: Slit width 0.681. You must hit the “Apply new values” button to activate the setting
      - Twin Secondary: Motorized Slit: Slit width 0.681. You must hit the “Apply new values” button to activate the setting
      - **Insert the Nickel K_β** filter on the secondary side of the optics

   c. Setup if using the **Göbel mirrors**:
      - Twin Primary: The Göbel mirror is more forgiving of poorly ground/defined samples (seems to give better S/N)
      - Twin Secondary: Motorized Slit: Slit width wide open at 5.4 mm. You must hit the “Apply to settings” button to activate the setting
      - **Insert a fixed slit** (standard is 0.1 mm) on the primary side or your peaks will be broad, and **remove the $K_\beta$filter** on the secondary side since the K_β radiation is already removed by the Göbel mirror

4) Initialize drives and x-ray tube:
   c. In “Commander” in the top left panel, anything with a caution symbol needs to be initialized. Click the checkbox to the right of each drive value and then click the “Initialized all checked drives” button, agree “yes”
   d. In the lower X-ray generator section, set Voltage to 40 kV and Current to 40 mA, hit the SET button

The system is ready for scanning

NOTE: *After 5 min in idle the system will ramp down the Voltage and Current to 20 kV and 5 mA*

Controlling the MTC-LOWTEMP Temperature Chamber (with everything ON)

1) In the Commander tab you should see a Temperature control panel below the X-ray generator panel
   a. Select whether you want just a straight temperature increase, or temperature with ramp (here you can adjust the ramping speed between 0-12 °C/min)
   b. Type in your desired temperature and ramp speed
   c. Hit the “Apply settings to instrument” button; the temperature should now start to ramp

2) When the stage has reached your desired temperature, wait until the temperature is stable (usually overshoots by 5° C), then run your XRD scans

Scan settings for D8 and managing your results

NOTE: *The numbers given below are just standard settings and you're free to change them to what you need. The numbers that you should* ***NOT*** *mess with are the x-ray power and current as well as the STEPS parameter.*

1) Choose scan parameters and start scan:
   a. If not done already, Initialize drives and x-ray tube (see above)
   b. In the "Scan setup" panel, fill in desired scan settings (see below for general settings); click into a white box several times to update the Effective total time [sec]; adjust dwell time and increments if total time does not fit your needs
   c. Hit START; the X-rays will turn on, power is ramped up, drives go to starting position and the measurement is started
   d. **After each scan, save your results file or the data will be lost!** When the scan has finished go to FILE > SAVE RESULTS ***both*** as .brml and .raw in a local folder and only then prepare next scan or sample

General Settings for Data Acquisition:

Sample rotation:	15 rpm
Time (s):	Your choice; more time equals better signal/noise (S/N)
2Theta angles:	Your choice; depends on your sample type
Increment (°):	Your choice; a smaller increment equals better S/N but you should set the increments so you get at least 5 data point above the half-height
Scan type:	Coupled two theta/theta
Scan mode:	Continuous PSD fast (Position Sensitive Detector)
PSD opening:	4.84876814600

NOTE: *Do* ***NOT*** *adjust the Steps parameter, it is automatically calculated from the other parameters. By adjusting steps improperly you can crash the optics!*

Ending your session and shutting down the D8

NOTE: *The .brml and .raw files that you have saved so far can only be opened with DIFFRAC.EVA and JADE; please follow the steps below to make your data accessible to you*

3) Data processing:
   a. Close DIFFRACT.measurement
   b. Import your .brml files in DIFFRAC.EVA and then export them as .xy file; this file extension can be opened in notepad, it will have two columns, 2Theta and CPS, which can be copied to excel.
   c. In DIFFRAC.EVA you can also search/match and append the matches if you like, and then print your spectrum.
   d. You can install DIFFRAC.EVA on your windows PC, just follow the instructions on the argenta2 XRD webpage (see link on top of first page); note that this software can only be used within the chemistry domain network
   e. If you want to refine your scans in Jade please ask Nina or Stephen to arrange Jade usage. The .raw file will be needed for that purpose.
   f. Copy saved files into your argenta2 folder (file browser Y: data \\argenta2.rd.unr.edu > Bruker D2) or on a thumb drive
   g. Log off from Windows

1) Turn off the temperature stage
   a. Set the temperature to 25° C and wait until the system reaches 25° C
   b. Once the system reaches 25° C turn off the heater on the Eurotherm (red button) then turn off the power to the Eurotherm (white button)
   c. Close the source and sink water lines on the floor behind the instrument
   d. Turn off vacuum pump (if used)
   e. Turn off vacuum gauge (if used)
   f. Turn of compressed air on the wall above the Haskris chiller (if used)

The Temp stage is now fully turned off

4) Turn off the D8 instrument:
   a. Press the generator button on the top left-hand front of the instrument (after a few seconds the button icon will change from a radiation symbol to a white "I")
   b. Press the white "Stand by" button in the insert on the lower left hand side of the instrument
   c. Turn the big yellow-red Mains knob from I to O (next to the Stand by button)

5) Stop the cooling cycle:
   a. Switch off the Haskris chiller
   b. Close water mains (2 valves) on wall; handles should be perpendicular to lines

Keep UPS PowerVar ON except for UPS maintenance

Installation of MTC-LOWTEMP Temperature Chamber

Temperature range: -180° to 450° C
For temperatures < 25° C or temperatures > 300° C vacuum is required
For temperatures < 25° cooling with LN_2 is required

6) Before switching stages make sure the goniometer arms are out of the way by reinitializing the drives (Commander, top checkbox, Initialized all checked drives)
7) Shut down the computer:
   a. Exit DIFFRAC.measurement (some settings in "Commander" tab will be lost)
   b. Open Measurement Server (task bar, right side), hit "Release control", then close window
   c. Shut down computer

****** NOTE: you must turn off the mains power to the instrument before removing a stage, if not you will damage the driver board ******

8) Turn off the instrument:
   d. Press the generator button on the top left-hand front of the instrument (after a few seconds the button icon will change from a radiation symbol to a white "I")
   e. Press the white "Stand by" button on the lower left hand side of the instrument
   f. Turn the Mains knob from I to O (next to the stand by button)

9) Remove the currently attached stage by following the instructions in the respective SOP

10) Attach the Temp stage:
   a. Using a 4 mm wrench remove the screws from the storage stage
   b. Lay the stage on its face and insert the high tension power cables in the cylindrical brass plugs, use a 2 mm wrench to loosen and tighten the terminals of the cables
   c. Align the red dots on top of the goniometer and on the Temp stage goniometer flange, push goniometer flange in, turn whole flange clockwise 45 degrees
   d. Connect the safety circuit cable (silver)
   e. Connect the water lines (usually blue and yellow coming from the bottom of the instrument are connected together): white – white; yellow – yellow; blue – blue
   f. Connect green thermocouple
   g. Connect black line (vacuum sensor)
   h. Tighten the single centering screw on the goniometer flange to the left of the center; this centers the stage to the goniometer
   i. Using a T25 wrench replace the outer 3 retaining screws on the goniometer (twelve o'clock, four o'clock, and eight o'clock positions): ***note*** the screw at twelve o'clock is longer than the other two
   j. Release the centering screw

The Temp stage is now fully attached. The status of the stage can be viewed in DIFFRAC.measurement > DAVINCI tab